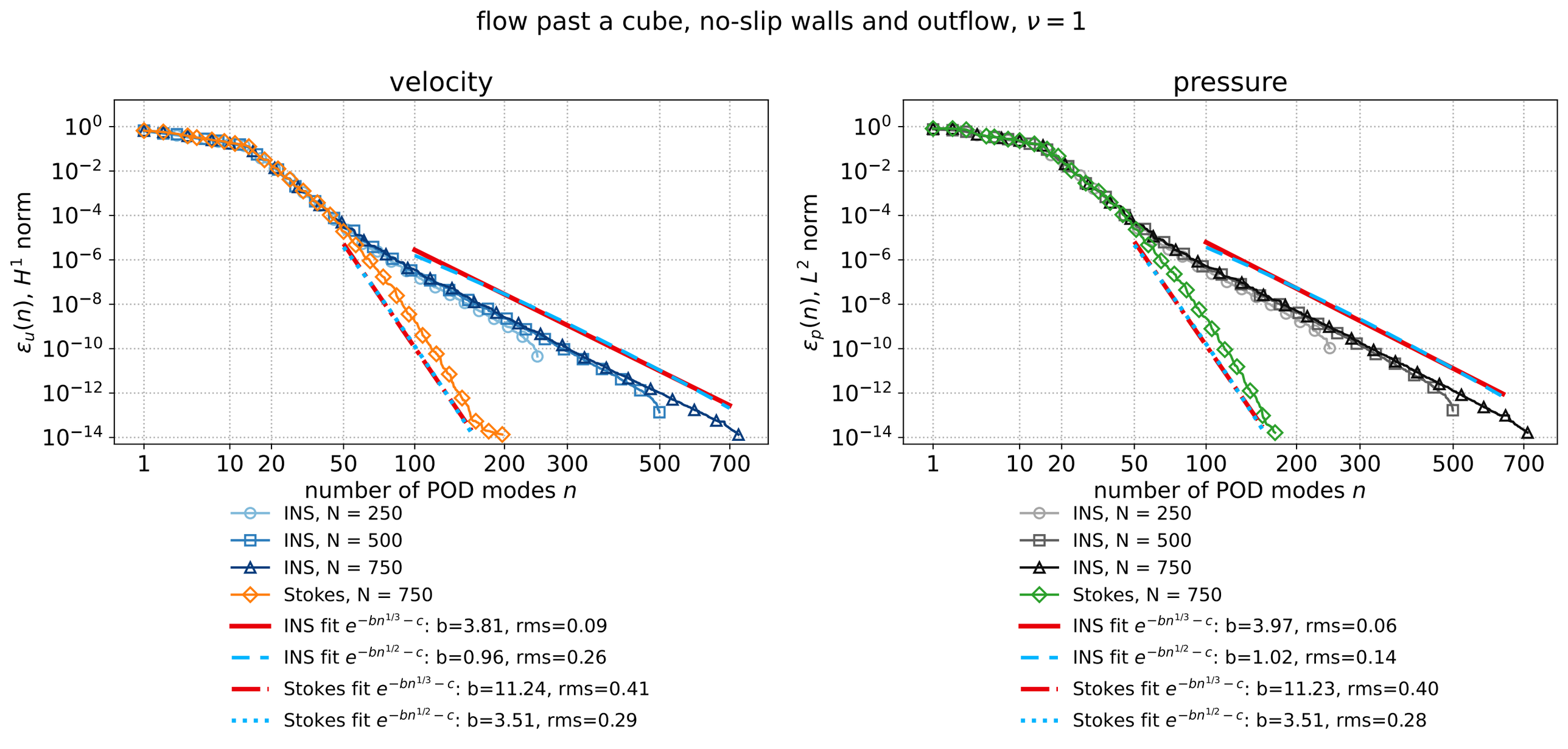

flow past a cube, no-slip walls and outflow, $\nu = 1$
velocity
pressure
$\varepsilon_u(n)$, H^1 norm
$\varepsilon_p(n)$, L^2 norm
number of POD modes n
number of POD modes n
INS, N = 250
INS, N = 500
INS, N = 750
Stokes, N = 750
INS fit $e^{-bn^{1/3}-c}$: b=3.81, rms=0.09
INS fit $e^{-bn^{1/2}-c}$: b=0.96, rms=0.26
Stokes fit $e^{-bn^{1/3}-c}$: b=11.24, rms=0.41
Stokes fit $e^{-bn^{1/2}-c}$: b=3.51, rms=0.29
INS, N = 250
INS, N = 500
INS, N = 750
Stokes, N = 750
INS fit $e^{-bn^{1/3}-c}$: b=3.97, rms=0.06
INS fit $e^{-bn^{1/2}-c}$: b=1.02, rms=0.14
Stokes fit $e^{-bn^{1/3}-c}$: b=11.23, rms=0.40
Stokes fit $e^{-bn^{1/2}-c}$: b=3.51, rms=0.28